

Computer Vision

How Machines See

From Pixels to Object Detection

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The Story So Far

Week	We Learned
5	Neural networks: MLPs, backprop, PyTorch
6	How to apply neural networks to IMAGES

A Child vs. A Computer

A 3-year-old child can:

- Recognize their parent in any photo
- Spot a dog across the street
- Know if a picture is upside down

A computer sees:

- Just numbers (pixels)
- No concept of "dog" or "parent"
- $224 \times 224 \times 3 = 150,528$ values

Today: How do we bridge this gap?

Why Computer Vision Matters

Application	What CV Does	Impact
Self-driving cars	Detect pedestrians, signs	Safety
Medical imaging	Find tumors, diagnose	Healthcare
Instagram filters	Face detection, AR	Entertainment
Security cameras	Person tracking	Safety
Quality control	Defect detection	Manufacturing
Agriculture	Crop disease detection	Food security

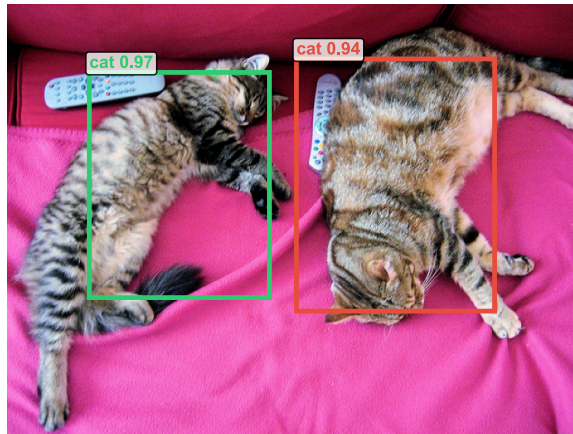
The Big Picture: Vision Tasks

Classification
"What is in this image?"

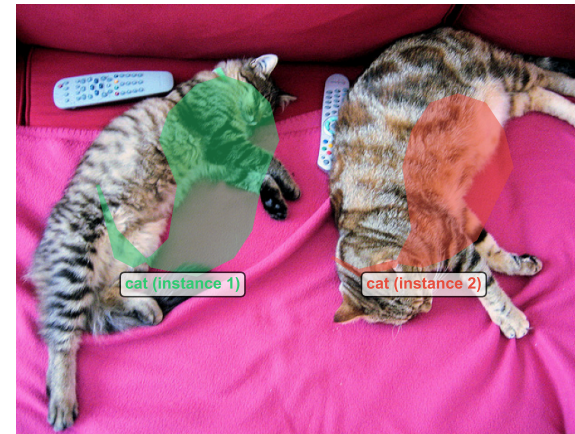


Label: "cat"

Object Detection
"What + Where?"



Instance Segmentation
"Precise boundaries (COCO masks)"



Today's Agenda

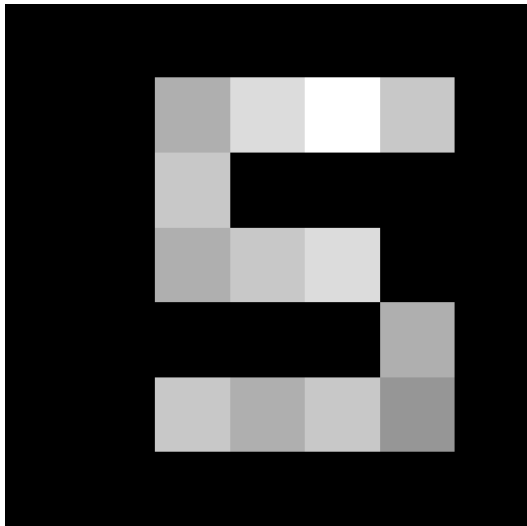
1. **Images as Data** - How computers "see"
2. **Convolution: The Key Insight** - Why it works
3. **CNN Architecture** - Building blocks
4. **From Classification to Detection** - The progression
5. **Object Detection** - What + Where
6. **YOLO** - Real-time magic

Part 1: Images as Data

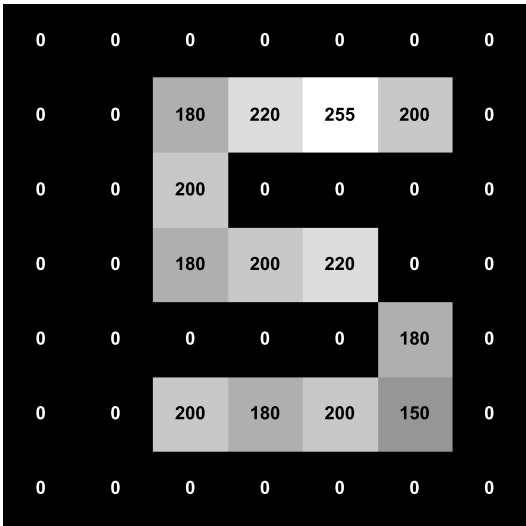
How Computers "See"

What Is an Image to a Computer?

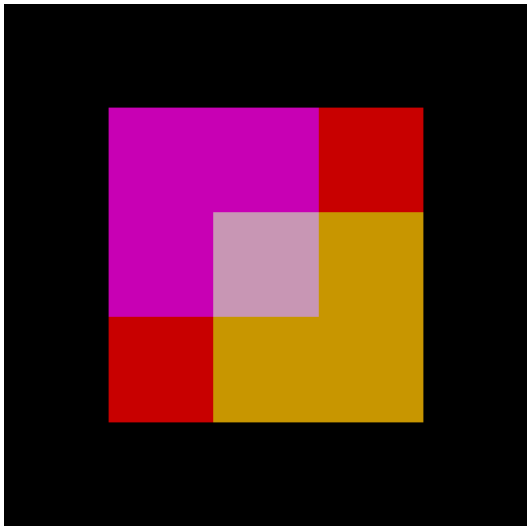
What We See
"The digit 5"



What Computer Sees
Just numbers (0-255)



Color Image = RGB Channels
3 numbers per pixel



Each pixel: [R, G, B] e.g. [200, 150, 180]

Grayscale vs Color

Type	Dimensions	Example
Grayscale	Height × Width	$28 \times 28 = 784$ values
Color (RGB)	Height × Width × 3	$224 \times 224 \times 3 = 150,528$ values

RGB = Red, Green, Blue channels

```
img = np.array(Image.open('cat.jpg'))  
print(img.shape)  # (224, 224, 3) → height, width, channels  
print(img[100, 200])  # [142, 87, 34] → R, G, B values
```

MNIST: The "Hello World" of Vision

Each MNIST digit is a 28×28 grayscale image = 784 pixels

Property	Value
Image size	28 × 28
Channels	1 (grayscale)
Pixel range	0 - 255
Total pixels	784

Task: Given 784 numbers, predict which digit (0-9)

Image Representation in Python

```
import numpy as np
from PIL import Image

# Load an image
img = Image.open('cat.jpg')
pixels = np.array(img)

print(pixels.shape)  # (480, 640, 3)
#                      Height, Width, RGB

# What does a pixel look like?
print(pixels[100, 200])  # [142, 87, 34]
#                      Red=142, Green=87, Blue=34 (brownish)

# Normalize for neural networks
pixels_normalized = pixels / 255.0  # Scale to 0-1
```

Quiz: What's the Difference?

Image A: A photo of a dog

Image B: The same photo, but flipped horizontally

To a human: Same dog!

To a computer: **Completely different numbers!**

```
np.array_equal(img_A, img_B) # False!
```

This is the fundamental challenge: same concept, different pixels.

The Three Challenges

Challenge	Why It's Hard	What We Need
Too many pixels	$224 \times 224 \times 3 = 150\text{K}$ features	Efficient architecture
Position doesn't matter	Cat on left = Cat on right	Translation invariance
Local patterns matter	Edges, textures, shapes	Local feature detection

Why Not Use MLPs for Images?

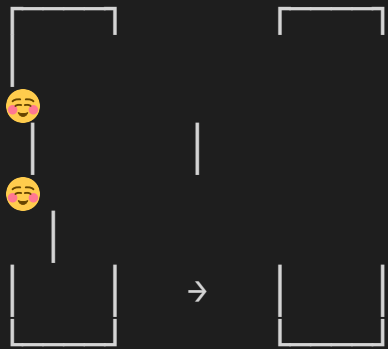
Problem 1: Too many parameters

- Image: $224 \times 224 \times 3 = 150,528$ inputs
- Hidden layer: 1,000 neurons
- Parameters: $150,528 \times 1,000 =$ **150 million weights!**

Why Not Use MLPs for Images? (cont.)

Problem 2: No spatial awareness

Original: Shifted:



MLP sees completely different pixels!

The same object at different positions looks completely different to an MLP.

What We Need

Requirement	Solution
Fewer parameters	Weight sharing
Translation invariance	Local patterns
Spatial structure	Keep 2D arrangement

Enter: Convolutional Neural Networks (CNNs)

Part 2: Convolutional Neural Networks

The Breakthrough (2012)

The Key Insight

Instead of connecting every pixel to every neuron...

Connect small **local regions** to neurons!

Image:

x	x	x	.	.	.
x	x	x	.	.	.
x	x	x	.	.	.
.	.	.	.	.	.

Filter:

w	w	w
w	w	w
w	w	w

*

= one output

3×3 filter

Convolution Operation

Slide a small filter across the image:

Image Filter Output

$$\begin{bmatrix} 1 & 2 & 3 & 0 & 1 \\ 0 & 1 & 2 & 3 & 2 \\ 1 & 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 3 & 2 \\ 2 & 1 & 0 & 1 & 2 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = [12, \dots]$$

$$1 \times 1 + 2 \times 0 + 3 \times 1 + 0 \times 0 + 1 \times 1 + 2 \times 0 + 1 \times 1 + 2 \times 0 + 1 \times 1 = 12$$

Why Convolution Works

Property	Benefit
Local connectivity	Each neuron sees a small region
Weight sharing	Same filter applied everywhere
Translation equivariance	Object detected regardless of position

Weight Sharing: The Big Win

Without sharing (MLP):

- $150,528 \text{ inputs} \times 1000 \text{ hidden} = 150 \text{ million weights}$
- Each position has different weights

With sharing (CNN):

- $3 \times 3 \text{ filter} \times 3 \text{ channels} \times 32 \text{ filters} = 864 \text{ weights!}$
- Same filter slides across entire image

```
150,000,000 vs 864 parameters  
(MLP)          (CNN)
```

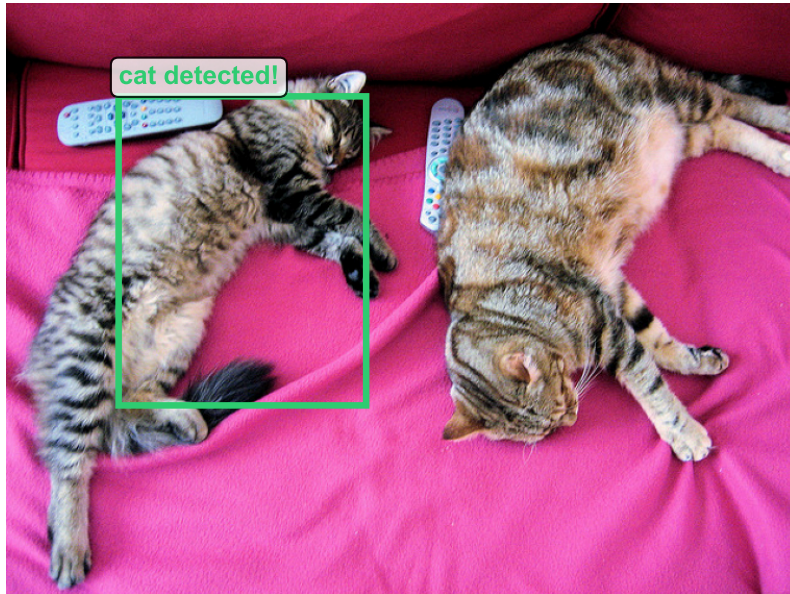
```
That's 173,000× fewer parameters!
```

Translation Equivariance

If the input shifts, the output shifts the same way:

Translation Equivariance: Same filter detects cat anywhere!

Original Position
Cat in left region



Different Position
Same cat, same filter works!



Same filter detects the cat regardless of position!

Filters Learn Features

Different filters detect different patterns:

Filter	What It Detects
Edge detector	Boundaries between regions
Corner detector	Sharp corners
Texture detector	Repeating patterns

The network **LEARNS which filters are useful!**

Convolution in Code

```
import torch.nn as nn

# Convolutional layer
conv = nn.Conv2d(
    in_channels=3,      # RGB input
    out_channels=32,    # 32 filters
    kernel_size=3,      # 3x3 filters
    padding=1           # Keep same size
)

# Apply to image
x = torch.randn(1, 3, 224, 224) # [batch, channels, H, W]
out = conv(x) # [1, 32, 224, 224]
```

Feature Maps

Each filter produces a **feature map**:

Input Image
(3 channels)



*

32 Filters
(3×3 each)



...

=

32 Feature Maps
(one per filter)



× 32

Part 3: CNN Architecture

The Building Blocks

CNN Building Blocks

Layer	Purpose	Output Change
Conv	Detect features	Different channels
ReLU	Non - linearity	Same size
Pooling	Reduce size	Smaller spatial
Flatten	2D $\rightarrow$ 1D	Vector
Linear	Classification	Class scores

Max Pooling

Take the maximum value in each region:

Input (4×4)
Colored 2×2 regions

1	3	2	1
4	2	6	5
7	8	1	0
3	5	9	2

Max Pool
2×2, stride 2

→

Take MAX
from each
region

Output (2×2)
Max from each region

4	6
8	9

Reduces spatial size by 2×

Why Pooling?

Benefit	Explanation
Reduces parameters	Smaller feature maps
Provides invariance	Small shifts don't matter
Increases receptive field	Later layers "see" more

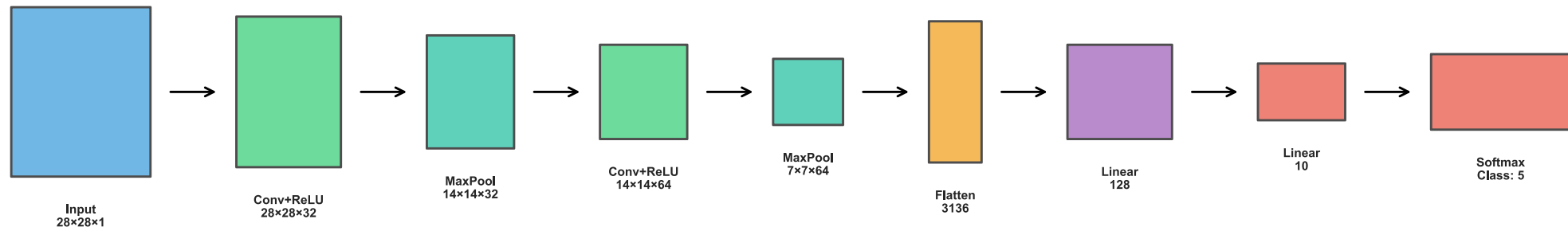
A Simple CNN

```
class SimpleCNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(1, 32, 3, padding=1) # 28→28
        self.pool1 = nn.MaxPool2d(2) # 28→14
        self.conv2 = nn.Conv2d(32, 64, 3, padding=1) # 14→14
        self.pool2 = nn.MaxPool2d(2) # 14→7
        self.fc = nn.Linear(64 * 7 * 7, 10) # Classify

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = self.pool1(x)
        x = F.relu(self.conv2(x))
        x = self.pool2(x)
        x = x.view(-1, 64 * 7 * 7) # Flatten
        x = self.fc(x)
        return x
```

CNN for MNIST

CNN Pipeline: Image $\rightarrow$ Features $\rightarrow$ Classification



Famous CNN Architectures

Year	Model	Key Innovation
1998	LeNet	First CNN
2012	AlexNet	Deep CNN, GPU training
2014	VGG	Very deep (16-19 layers)
2015	ResNet	Skip connections (152 layers!)
2017	EfficientNet	Neural architecture search

AlexNet: The Game Changer

ImageNet 2012:

Previous best (hand-crafted):	25.8%
AlexNet (CNN):	16.4%
Human performance:	~5%

← 9% improvement!

This started the deep learning revolution!

Transfer Learning

Don't train from scratch! Use pre-trained models:

```
import torchvision.models as models

# Load pre-trained ResNet (trained on ImageNet)
model = models.resnet50(pretrained=True)

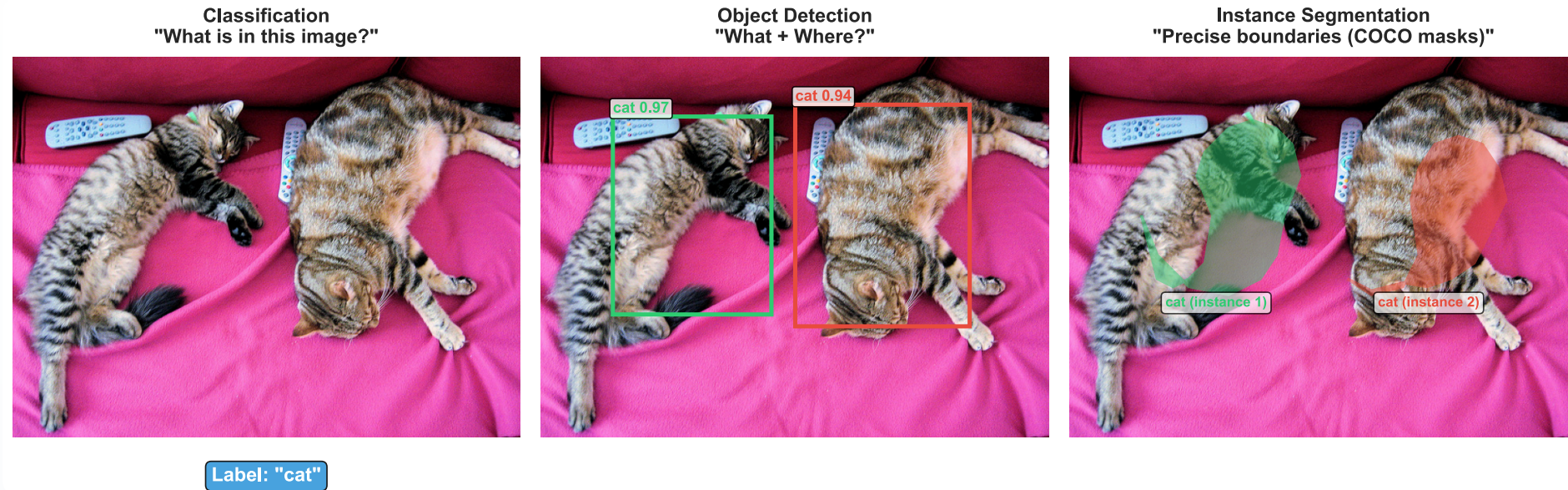
# Replace last layer for your task
model.fc = nn.Linear(2048, num_classes)

# Fine-tune on your data
```

Part 4: From Classification to Detection

The Journey from "What" to "What + Where"

The Progression of Vision Tasks



Classification → Localization → Detection

Each step adds more information about the objects in an image.

Classification vs Localization vs Detection

Task	# Objects	Output	Example
Classification	1 (assumed)	Class label	"This is a cat"
Localization	1	Class + 1 box	"Cat at (x,y,w,h)"
Detection	0 to many	Classes + boxes	"Cat at..., Dog at..."

Detection = Classification + Localization + Handle multiple objects

Why Detection is Harder

Challenge	Why It's Difficult
Variable count	0, 1, 5, or 100 objects in an image
Variable sizes	Same object can be huge or tiny
Overlapping objects	Objects can occlude each other
Background clutter	Must ignore non-objects
Real-time requirement	Self-driving cars need 30+ FPS

Why Detection Matters

Self-driving car needs to know:

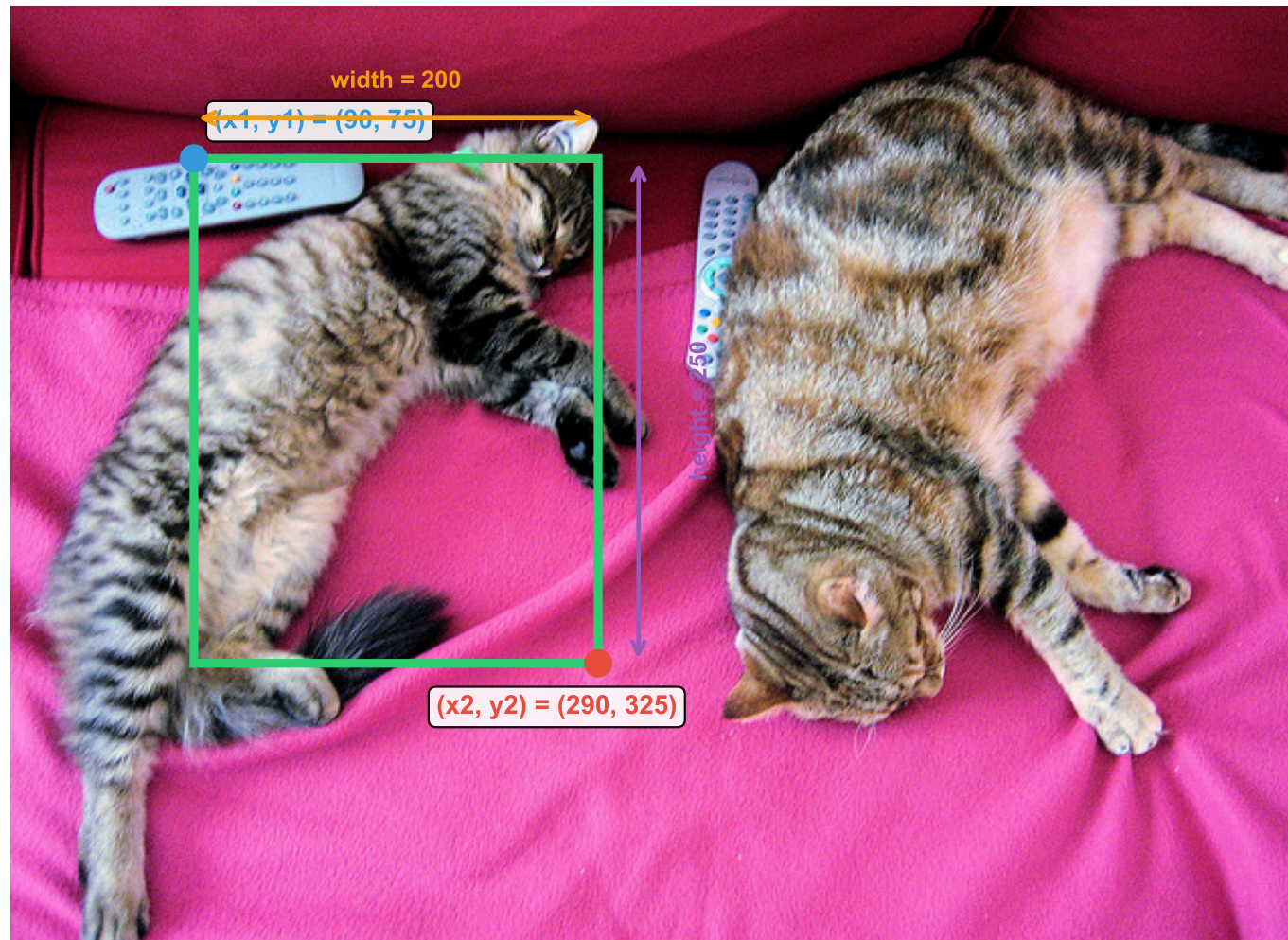
- What is there? (pedestrian, car, sign)
- Where is it? (position, distance)
- How many? (one child or five?)

Classification can't answer "where"!

Bounding Boxes

A bounding box = 4 numbers describing a rectangle:

Bounding Box: 4 Numbers Define Object Location
Format: (x1, y1, width, height) or (x1, y1, x2, y2)



Detection Output

For each object:

- **Class label:** "dog", "car", "person"
- **Bounding box:** (x1, y1, x2, y2)
- **Confidence:** 0.95 (95% sure)

```
detections = [  
    {"class": "dog", "box": [100, 50, 300, 200], "conf": 0.95},  
    {"class": "cat", "box": [400, 100, 550, 250], "conf": 0.87},  
]
```

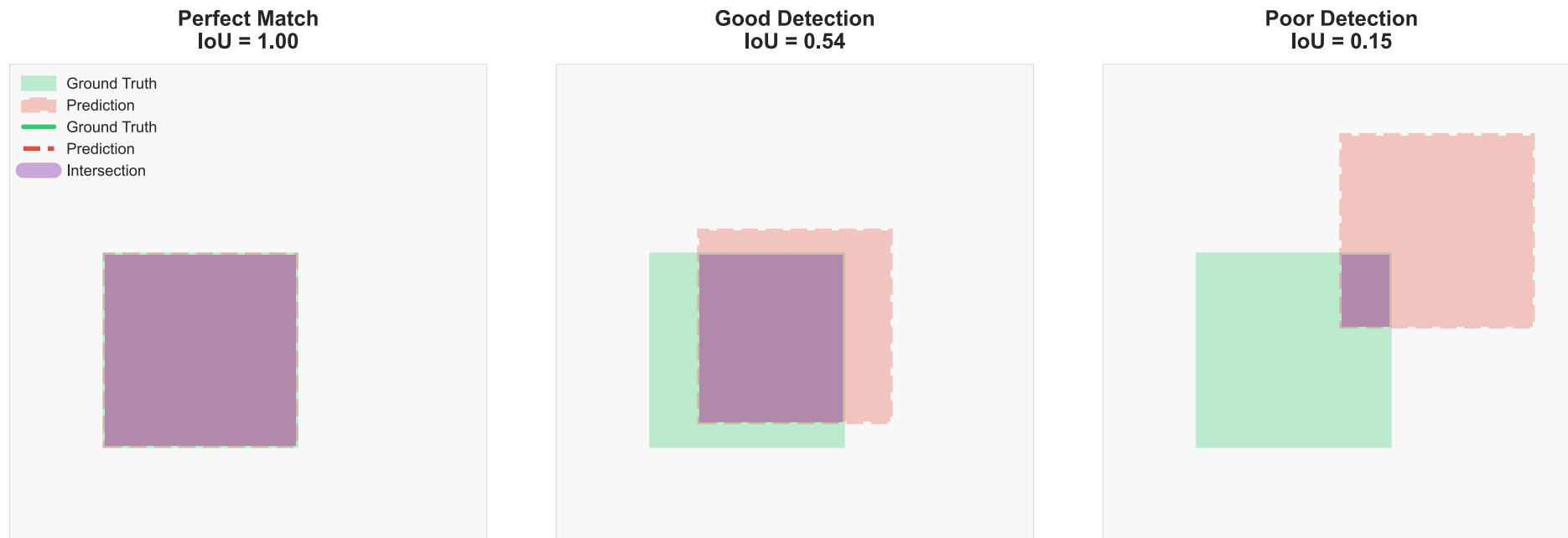
IoU: Measuring Box Overlap

Intersection over Union - How good is a predicted box?

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

IoU Value	Quality
1.0	Perfect match
> 0.5	Good detection
< 0.3	Poor detection

IoU Examples



IoU Thresholds

IoU Threshold	Meaning	Use Case
IoU ≥ 0.5	"Good enough"	Standard detection (PASCAL VOC)
IoU ≥ 0.75	"Precise"	High-quality detection
IoU ≥ 0.95	"Near perfect"	Fine-grained applications

mAP@0.5:0.95 averages performance across multiple IoU thresholds.

Computing IoU in Python

```
def compute_iou(box1, box2):  
    """  
    Boxes are [x1, y1, x2, y2] format  
    """  
  
    # Intersection coordinates  
    x1 = max(box1[0], box2[0])  
    y1 = max(box1[1], box2[1])  
    x2 = min(box1[2], box2[2])  
    y2 = min(box1[3], box2[3])  
  
    # Areas  
    intersection = max(0, x2-x1) * max(0, y2-y1)  
    area1 = (box1[2]-box1[0]) * (box1[3]-box1[1])  
    area2 = (box2[2]-box2[0]) * (box2[3]-box2[1])  
    union = area1 + area2 - intersection  
  
    return intersection / union if union > 0 else 0
```

Non-Maximum Suppression (NMS)

Problem: Model often detects the same object multiple times.

Solution: Keep only the best detection, remove overlapping ones.

```
# Many boxes for same dog
boxes = [[100,50,300,200], [105,55,305,205], [110,48,295,198]]
confs = [0.95, 0.92, 0.88]

# After NMS: keep only [100,50,300,200] with conf 0.95
```

Part 5: YOLO

You Only Look Once

YOLO: The Revolution

Before YOLO (2015): Detection was SLOW (seconds per image)

After YOLO: **Real-time detection** (30-60 FPS!)

Approach	Speed	How
R-CNN	~50 sec	Propose regions, classify each
Fast R-CNN	~2 sec	Share computation
YOLO	~0.02 sec	One network, one pass

YOLO Key Idea

Process the entire image in one forward pass:

1. Divide image into grid (e.g., 7×7)
2. Each cell predicts:
 - Bounding boxes
 - Confidence scores
 - Class probabilities
3. Single neural network does everything!

YOLO Grid

YOLO: 7×7 Grid Overlay on Real Image
Highlighted cells are responsible for detecting objects



Using YOLO in Practice

```
from ultralytics import YOLO

# Load pre-trained model
model = YOLO('yolov8n.pt')

# Detect objects
results = model('image.jpg')

# Process results
for result in results:
    for box in result.boxes:
        x1, y1, x2, y2 = box.xyxy[0].tolist()
        conf = box.conf[0].item()
        cls = int(box.cls[0].item())
        print(f"Class {cls}: ({x1:.0f},{y1:.0f}) to ({x2:.0f},{y2:.0f})")
```

YOLO Model Sizes

Model	Speed	Accuracy	Use Case
YOLOv8n	Fastest	Lower	Mobile, edge
YOLOv8s	Fast	Medium	General use
YOLOv8m	Medium	Good	Better accuracy
YOLOv8l	Slower	Better	High accuracy
YOLOv8x	Slowest	Best	Maximum accuracy

Training Your Own Detector

```
from ultralytics import YOLO

# Start with pre-trained model
model = YOLO('yolov8n.pt')

# Train on custom data
model.train(
    data='my_dataset.yaml',
    epochs=100,
    imgsz=640
)

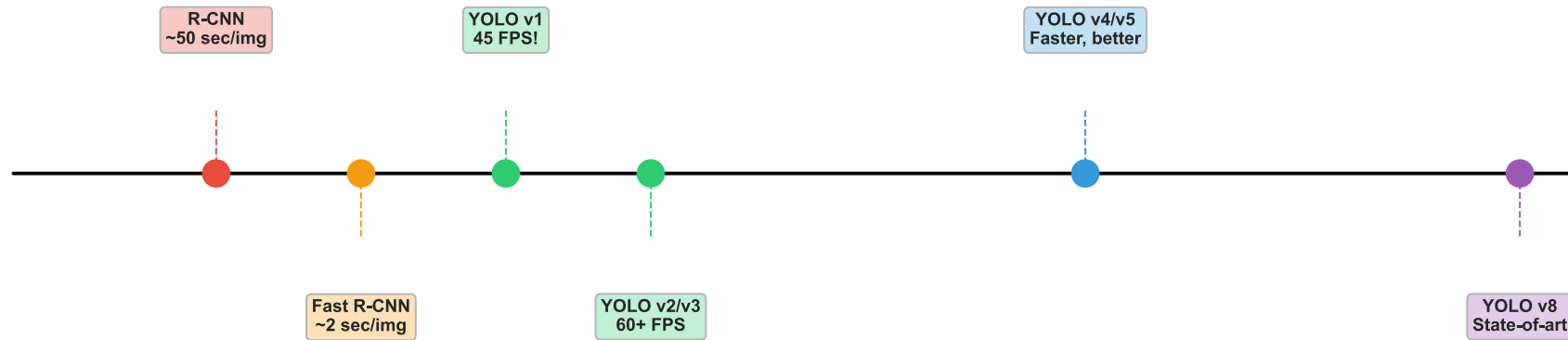
# Evaluate
metrics = model.val()
print(f"mAP@0.5: {metrics.box.map50:.3f}")
```

Detection Metrics

Metric	What It Measures
Precision	Of detections, how many correct?
Recall	Of actual objects, how many found?
mAP@0.5	Average precision at IoU=0.5
mAP@0.5:0.95	Average over IoU thresholds

Detection Speed Evolution

Evolution of Object Detection
From Minutes to Milliseconds



Real-World Detection Applications

Application	Objects Detected	Speed Requirement
Self-driving cars	Cars, pedestrians, signs	Real-time (30+ FPS)
Retail analytics	People, products	Near real-time
Medical imaging	Tumors, lesions	Accuracy > speed
Sports analysis	Players, ball	Real-time
Security cameras	People, vehicles	Real-time
Manufacturing QC	Defects, parts	High accuracy

Beyond Detection: Segmentation

Instance Segmentation = Detection + Pixel-level masks

Task	Output	Use Case
Detection	Bounding boxes	"Where are objects?"
Semantic Seg.	Pixel classes	"What is each pixel?"
Instance Seg.	Pixel masks per object	"Which pixels belong to which object?"

Segmentation Variants Comparison

Semantic Segmentation
"All cats = same color"



Instance Segmentation
"Each cat = different color"



Panoptic Segmentation
"Things (countable) + Stuff (regions)"



Segmentation with YOLO

```
from ultralytics import YOLO

# Load segmentation model
model = YOLO('yolov8n-seg.pt')

# Run segmentation
results = model('image.jpg')

# Access masks
for result in results:
    masks = result.masks # Segmentation masks
    boxes = result.bboxes # Bounding boxes
```

Pose Estimation

Beyond "what" and "where" → "how is it positioned?"

Pose estimation detects keypoints (joints) on bodies:

- 17 keypoints for humans (COCO format)
- Enables gesture recognition, action detection
- Used in sports, healthcare, gaming

```
# Pose estimation with YOLO
model = YOLO('yolov8n-pose.pt')
results = model('person.jpg')

# Get keypoints
for result in results:
    keypoints = result.keypoints # Shape: (n_persons, 17, 3)
```


The Vision AI Stack

Layer	Components
Applications	Self-driving, Medical, Retail, Security, Robotics
High-Level Tasks	Pose Estimation, Tracking, Action Recognition
Core Tasks	Classification, Detection, Segmentation
Building Blocks	CNNs, Transformers (ViT), Feature Extractors
Fundamentals	Images as Tensors, Convolutions, Pooling

Each layer builds on the ones below it!

From Images to Video

Video = Sequence of Images (Frames)

Task	Description	Challenge
Object Tracking	Follow objects across frames	Maintain ID consistency
Action Recognition	Classify what's happening	Temporal understanding
Video Object Detection	Detect objects in video	Real-time processing

Tracking adds a new dimension: **TIME**

Multi-Object Tracking (MOT)

Challenge: Same dog appears in frame 1, 2, 3...
How do we know it's the SAME dog?

Solution: Track IDs

```
from ultralytics import YOLO

model = YOLO('yolov8n.pt')

# Track objects across video frames
results = model.track(source='video.mp4', persist=True)

# Each detection now has a track ID
for result in results:
    for box in result.boxes:
        track_id = box.id # Consistent across frames!
```

Key Takeaways

1. **Images = grids of numbers** (pixels)
2. **CNNs use convolutions** for efficient image processing
 - Weight sharing (fewer parameters)
 - Translation equivariance (detect anywhere)
3. **CNN building blocks**: Conv → ReLU → Pool → Flatten → Linear
4. **Detection = Classification + Localization**
 - Bounding boxes + class labels + confidence
5. **IoU** measures box overlap (threshold typically 0.5)
6. **YOLO** enables real-time detection (single forward pass)
7. **Beyond detection**: Segmentation, Pose, Tracking

Connecting to Next Week

Computer Vision answers: "What's in this image?"

Next question: "What does this text mean?"

This Week	Next Week
Images → Pixels → CNNs	Text → Tokens → Transformers
Object Detection	Next Token Prediction
YOLO (real-time vision)	LLMs (language understanding)

Same principle: Learn patterns from data to make predictions!

You Can Now Make Computers See!

Next: Language Models - How Machines Understand Text

Lab: Build a CNN classifier and run YOLO detection

"The question is not whether intelligent machines can have any emotions, but whether machines can be intelligent without any emotions."

— Marvin Minsky

Questions?